NewsBot Intelligence System

ITAI 2373 – Natural Language Processing

Student: Patrick Rostand Gandjouon Tchassem
Project: Midterm NewsBot Intelligence System

1. Introduction

The purpose of this project was to build a NewsBot Intelligence System that uses Natural Language Processing (NLP) to analyze news articles. NLP allows computers to process and understand human language.

In this project, I used different NLP techniques to clean text, identify important words, analyze sentence structure, detect sentiment, recognize named entities, and classify news articles. My goal was to understand how these techniques work together in a system that can organize and analyze news information.

2. Dataset and Preprocessing

I used the BBC News dataset, which contained 1,490 articles in five categories: Business, Entertainment, Politics, Sport, and Technology. After removing duplicates, 1,440 articles remained.

Before analyzing the articles, I cleaned the text by converting words to lowercase, removing unnecessary characters and stop words, and applying lemmatization. These steps helped make the text easier for the models to process.

I also compared the original and cleaned text to understand how preprocessing changes the information available for analysis.

3. NLP Techniques Used

TF-IDF Analysis: I used TF-IDF to identify important words in each article. The system created 5,000 text features. Different news categories showed different word patterns, which helped explain how the model could recognize article topics.

Part-of-Speech Tagging: I used POS tagging to identify nouns, verbs, and adjectives. Nouns were the most common grammatical category. This helped me understand how grammatical patterns can be analyzed across news categories.

Dependency Parsing: I used spaCy to identify relationships between words in sentences. The diagrams helped me understand sentence structure, although longer sentences were sometimes difficult to follow.

Sentiment Analysis: I used VADER to classify articles as positive, negative, or neutral. Out of 1,440 articles, 1,013 were positive, 416 were negative, and 11 were neutral. I learned that sentiment analysis is useful, but it does not always understand the full meaning of an article.

Named Entity Recognition: I used spaCy to identify people, organizations, locations, dates, and other important information. The system extracted 40,803 entity mentions. Some entities were incorrectly identified, showing that the results still need review.

4. Machine Learning Classification

I trained and compared three machine learning models to classify articles into news categories.

Model	Test Accuracy
Naive Bayes	93.75%
Logistic Regression	93.06%
Support Vector Machine	75.00%

Naive Bayes performed the best, achieving 93.75% test accuracy. I also used a confusion matrix to examine the predictions and identify classification errors.

These results showed that machine learning can classify news articles accurately, but some articles can be difficult to categorize when they discuss more than one subject.

5. NewsBot System Testing

After completing the NLP tasks, I combined them into one NewsBot system that could classify articles, analyze sentiment, extract named entities, and generate insights.

I tested the system using three example articles:

- **Microsoft AI acquisition:** Classified as Technology with 41.41% confidence.
- **Lakers championship:** Classified as Sport with 80.30% confidence.
- **Climate change report:** Classified as Business with 36.28% confidence.

All three articles were processed successfully.

The climate change article showed a limitation because the system did not have an environmental news category. Its sentiment was also labeled positive despite discussing serious concerns. This helped me understand why AI results should be reviewed carefully.

6. Challenges and Lessons Learned

One of my biggest challenges was combining the different NLP techniques into one working system. I experienced errors when the model received input that did not match the expected features.

I worked through these problems by reviewing the code, checking the data, and testing the system again.

I also learned that high accuracy does not mean a model will always make correct predictions. Testing new articles helped me understand the importance of checking results instead of relying only on accuracy scores.

7. Business Applications and Ethical Considerations

NewsBot could help media companies automatically organize news articles. Businesses could also use it to monitor news about companies, products, and events.

Sentiment analysis could help identify the tone of news coverage, while Named Entity Recognition could help find important people and organizations.

However, incorrect predictions and bias in training data could lead to misleading results. Privacy and responsible use of information are also important.

I would recommend improving the training data, adding more categories, reviewing low-confidence predictions, and keeping human review for important decisions.

8. Conclusion

This project helped me understand how different NLP techniques work together to analyze news articles.

I learned how to prepare text, extract important information, analyze sentiment, and train machine learning models. The best model achieved 93.75% accuracy, and the NewsBot system successfully processed three new articles.

Although the system had some limitations, this project helped me improve my understanding of NLP and how it can be used to solve real-world problems.